

Certified analytical bounds for hypersonic reachable sets

A two-page summary of a preliminary PhD research proposal

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Full proposal: research.ekdynamics.science/proposal.pdf

The result. This is a problem in nonlinear analysis: a singularly perturbed family of differential inclusions in which compactness fails along the time axis, because the motion carrying the important state transitions occupies a vanishing time. It therefore disappears from the ordinary limits of the family while continuing to decide where the trajectories go. For a model that keeps exactly that phenomenon I can prove

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\mathcal{R}_\mathcal{H}([T - \varrho, T + \varrho])}_{\text{skeleton, bubbles and glide}} \oplus \underbrace{\mathbb{B}(\varrho)}_{\text{residual}},$$

with the residual a sum of three terms — the drift dropped over the passes, the error of averaging the fast oscillations, and the air the ballistic arcs discard — each bounded in closed form by one estimate, and with every constant explicit in the data and free of ε . The inclusion is uniform over the controls, over the initial set and over a bounded family of admissible fields. One threshold sets the balance between the three, and at its best choice $\varrho = O(\varepsilon^{1/4})$; I claim the order and not its sharpness. The proof is two pages and elementary; the statement, the hypotheses and the proof are in the proposal. What follows says what the object is, why the decomposition is the natural one, and what remains open.

The problem. The structural phenomenon is concentration, and the language for it is a profile decomposition: a low-dimensional *skeleton*, one *bubble* per pass, and a *residual*. In the theory of nonlinear dispersive equations such a decomposition comes from iterating the dichotomy of the concentration-compactness principle [1, 2], and what it delivers is structure without a constant — by contradiction. The proposal is that here the structure can be had quantitatively, because the family carries Lyapunov functionals: the dissipation is monotone with total variation bounded uniformly in the parameter, so the number of profiles above any threshold is bounded in advance rather than extracted. That is where I expect the contribution to the analysis to lie.

The example that supplies the problem is hypersonic atmospheric entry, where the solution set is the footprint a vehicle can reach and the parameter is the ratio of the atmospheric scale height to the planetary radius. It is the example and the test, not the subject. Computing the set exactly means a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in the state dimension and capped in practice near four or five states [3], while a rigid-body entry vehicle carries thirteen. The dimension count is the most legible way to say what the decomposition earns, though it is not what finally binds: the bound is to be a pair of functions exhibited and checked rather than a mesh, and what binds is the enclosure of the jump map.

The model. The state is (z, v, s) , with z an altitude, v its rate and s a vector of slow coordinates, and

$$\dot{z} = v, \quad \dot{v} = c m^\varepsilon(z) - 1 - \kappa v, \quad \dot{s} = F + m^\varepsilon(z) \Gamma, \quad m^\varepsilon(z) = m(z/\varepsilon),$$

with m positive and strictly decreasing and c a scalar control: a unit-mass particle in the potential Φ_c with $\Phi'_c = 1 - c m^\varepsilon$, one soft wall and uniform gravity above it. The small parameter sits in one place only, the width of the band in which the air is felt; the equations of motion carry none. All of m, κ, F, Γ may be chosen by a selection from a bounded family, so the system is a differential inclusion and the density itself is enclosed rather than fixed.

The model is the entry oscillator rather than a caricature assembled for the occasion, and three regimes of flight are three energies $W = \frac{1}{2}v^2 + \Phi_c(z)$ in one well: the minimum is the equilibrium glide, a small oscillation is the phugoid, a large one is a skip. *The dip is short and the arc is long because the well is asymmetric*, and that asymmetry, read at the scale of the band, is the whole of the concentration. Two Lyapunov functionals do the quantitative work — the amplitude W for the fast motion and the dissipation for the slow — and three exact identities organise it: $\langle m^\varepsilon \rangle = 1/c$ at every amplitude, so the glide is not merely the bottom of the well but the *mean* of every oscillation about it; ε sets no shape, only a clock, but the clock is read at amplitude W/ε , which is why a large oscillation keeps a period of order one while its dip shortens like $\varepsilon \log(1/\varepsilon)$; and the impulse a pass delivers is twice its entry speed over c , so the dip takes a vanishing share of the time and essentially all of the effect. The limit system $\mathcal{H}$ flows ballistically on the arcs, jumps at $z = 0$ by a map J — the solution set of an inner problem carrying no small parameter — and flows by the averaged field $F + \Gamma/c$ on the glide. The hypotheses, the constants and the two-page proof are in the proposal.

What makes the proof elementary is that the functionals do all the counting. They bound the number of passes above any fixed size, with no ε in the bound; they bound the summed duration of the dips, and the time the motion may spend in closed orbits; and they bound the accumulated amplification, which is exponential in an impulse and therefore bounded however many passes there are. The one discipline throughout is that the density, which is unbounded inside the band as $\varepsilon \rightarrow 0$, is never bounded pointwise there but always by its impulse — which is exactly

what the dissipation functional controls. No compactness argument enters. That is the point rather than an evasion: the structure is the one concentration-compactness describes, and the proposal is that a Lyapunov functional makes it quantitative where the classical dichotomy makes it only qualitative.

Where the measures enter. The theorem hands the occupation-measure programme of Lasserre, Henrion and collaborators [4] a different problem. $\mathcal{R}_{\mathcal{H}}$ is a reachability problem in $2 + d$ coordinates with one guard and a set-valued reset, which is the hybrid form for which the relaxation is known [5]: the Liouville equation is a linear constraint on a measure, a measure on the guard carries the mass across each pass, and the dual certificate is a pair of functions non-decreasing along the flow and across the reset. The singular limit is discharged by the theorem, which is elementary, and what is left on $\mathcal{H}$ is a problem in a few coordinates. Whether the limit is better taken in this topology, with the decomposition extracted from the measures, or at the level of trajectories as in the proof, is open — and the two routes are not known to be equivalent.

What remains. Every hypothesis of the theorem holds for entry. What does not hold is that its residual is small at the physical ε . A pass entered at flight-path angle γ_e enters through the ratio $\gamma_e/\sqrt{\varepsilon}$, as in the classical Chapman variables [6], and for the Earth $\sqrt{\varepsilon} \approx 0.034$ is about two degrees while entry angles are a few degrees. So the amplitude at which a dip becomes brief is the amplitude at which the period becomes long, and no threshold makes both approximations accurate at once. Four problems follow.

1. **The decomposition itself.** That every sequence of trajectories splits, along a subsequence, into the skeleton, finitely many profiles obtained by rescaling the passes to their own widths, and a remainder small in a stated norm. The theorem above compares a trajectory with a trajectory of the limit system and does not assert this. What the classical argument gets by contradiction and without a constant — that the profiles above a given size are finite in number — the dissipation functional here gives in advance and explicitly, and that bound is already proved; what is open is the decomposition it would organise.
2. **The statement at a fixed ε .** The orbits near the threshold, which neither approximation serves, must be carried by the unapproximated field. That adds no term to the residual, because nothing is approximated; it costs dimension, over a total duration the budget bounds. What has to be shown is a bound on the growth of that enclosure, explicit in the interval widths — and, with it, the smaller selection class the real atmosphere permits, which is where the bound would most sharpen.
3. **Separation of scales, by enclosure.** The attitude and elastic motions do not concentrate, and averaging them is the obvious route and the wrong one: an averaging theorem gives an order and not a bound, and holds through a resonance only outside a set of small measure, which an outer bound cannot discard. They act on the trajectory only through the aerodynamic force, so it is the force that must be enclosed. Two facts are to be established: where there is no air there is no torque, and the attitude has no aerodynamic effect in any case; and where there is air the damping rate is proportional to density times speed, while the energy identity fixes the integral of density against the cube of speed, so the budget that bounds the dissipation also bounds the integrated damping from below.
4. **The companion at the level of measures.** That the limit of the occupation measures satisfies the Liouville equation of the limit system, with the mass crossing each pass carried on the graph of J , is a separate statement from the one above, and it is where the concentration machinery is expected to be needed — for the structure rather than for the constants.

The first and fourth are the analysis; the second and third carry the model to the application. The fourth is the question I would most value a conversation about: granting the limit of the measures and the profiles it carries, whether the limiting solution set is contained in the one those profiles generate is a further implication, and the hardest link in the chain. It is the part of the programme for which I am looking for an advisor.

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